

Training Institute For Chemical Industries

Polash, Narsingdi, Bangladesh

(Central Training Institute of BCIC)



Certificate

This is to certify that

Md. Hamed Hasan

has successfully completed the training course on

Chemical Analysis &

Quality Control

of 03 (Three) weeks duration conducted by this Institute

from 13.07.2024 to 01.08.2024

*He/She obtained grade **A** in the final evaluation.*

Date: 01 August, 2024

Executive Director

Gradation Criteria :

A⁺ = 80% & above, A = 75 ~ <80%, A⁻ = 70 ~ <75%, B⁺ = 65 ~ <70%, B = 60 ~ <65%
B⁻ = 55 ~ <60%, C⁺ = 50 ~ <55%, C = 45 ~ <50%, D = 40 ~ <45%, F = Below 40%

Course Outline

Analytical Chemistry & Environmental Science Department (60 Hours):

1.0 Quality products and productivity

Introduction to quality control and quality assurance.

2.0 Analysis of industrial water

Determination of Na & K in green coconut by Flame Photometer; Physical parameters- pH, Conductivity, Turbidity, TDS.

3.0 Quality testing methods in process industries

Preparation & standardization of solutions for titrimetric analysis; Estimation of caffeine in tea/ coffee/ cola drink; Viscosity measurement of thickening agent; Determination of total sugar in beverage using polarimeter.

4.0 Quality testing methods in food industries and pharmaceuticals

Test of Adulteration; Estimation of fat in milk / ice cream / powder milk; Analysis of pharmaceutical products- metronidazole/metformin; Estimation of Vitamin A in edible oil; Estimation of antioxidant: ascorbic acid and pectin content in fruits; Estimation of Polyphenol/ carotene in fruit and vegetables.

5.0 Spectrometric Methods of Analysis

Principles and instrumentation of spectrophotometer; Identification & Quantification of analyte by UV/Vis Spectrophotometer; Analysis of ciprofloxacin tablet by UV-Vis. Spectrophotometer; Qualitative soil micronutrient analysis by AAS / ICP-OES; Quantitative soil micronutrient analysis by AAS / ICP-OES; Identification of packaging materials by FTIR; Operation and application of EDXRF.

6.0 Chromatographic Methods of Analysis

Principles and instrumentation of chromatograph; Chemical analysis by GC-TCD and data interpretation; Chemical analysis by GC-FID and data interpretation; Operation and application of GC-MS; Estimation of caffeine in cola drinks by HPLC.

7.0 Environmental pollution monitoring and control

Wastewater treatment techniques; Determination of COD in Wastewater; Measurement of pollution parameters: DO, BOD in water.

Industrial Safety & Health Department (06 Hours):

8.0 Safety measures in process industry

Introduction to industrial safety; Fire hazards and fire protection arrangement; Explosion hazards and personal protective equipment.

Administration and Accounts & Finance Departments (04 Hours):

9.0 General

National Integrity Strategy (NIS); Accounts and Finance in Industrial Sector.